

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

Theory University Examination March-April 2018

PH418 Medicinal Chemistry-III

Date: 26.3.2018, Monday Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 80

Instructions:

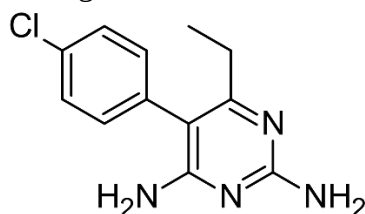
1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. The following structure is of _____



- [A] Trimethoprim
[B] Pyrimethamine
[C] Primaquine
[D] Halofanterine
2. One of the following is consist of imidazole moiety in its structure.
[A] Ciclopirox
[B] Butaconazole
[C] Griseofulvin
[D] Cotrimoxazole
3. The mechanism of action of rifampin includes an inhibition of which of the following enzymes?
[A] DNA dependent RNA-polymerase
[B] Transpeptidase
[C] Topoisomerase II
[D] RNA-dependent DNA-polymerase
4. Which of the following drug is not inhibiting ergosterol synthesis?
[A] Nystatin
[B] Butenafine
[C] Ketoconazole
[D] Terbinafine

5. Which of the following antiviral drugs are not thymidine or deoxythymidine analogue?
 - [A] Stavudine
 - [B] Telbivudine
 - [C] Vidarabine
 - [D] Zidovudine
6. Which of the following statement is not true for cyclophosphamide-an anticancer drug?
 - [A] It is relatively non toxic
 - [B] It acts as a prodrug
 - [C] It cannot be taken orally
 - [D] The structure is metabolized to acrolein.
7. What are the two main targets currently used in anti-HIV therapy?
 - [A] Reverse transcriptase and protease
 - [B] Reverse transcriptase and integrase
 - [C] Protease and integrase
 - [D] The viral glycoproteins gp120 and gp41
8. Combination of allopurinol with this anticancer drug can cause an increase in toxicity
 - [A] 6-thioguanine
 - [B] Fluorouracil
 - [C] 6-mercaptopurine
 - [D] Azatadine
9. 1, 3-Diamino cyclohexane central ring absent in _____.
 - [A] Kanamycin
 - [B] Gentamicin
 - [C] Tobramycin
 - [D] Streptomycin
10. Streptomycin is obtained from _____.
 - [A] *Micromonospora purpurea*
 - [B] *Streptomyces griseus*
 - [C] *Streptomyces rimosus*
 - [D] *Streptomyces tenebrarius*
11. Diloxanide is not useful in _____.
 - [A] Intestinal amoebiasis
 - [B] Cyst passers
 - [C] Extraintestinal amoebiasis
 - [D] Mixed amoebicides
12. Chlorhexidine _____ antiseptic agents.
 - [A] Amidines
 - [B] Furan derivatives
 - [C] Biguanides
 - [D] Aldehydes
13. Which of the following cause inhibition of protein synthesis?
 - [A] Polymyxin
 - [B] Linezolid
 - [C] Ciprofloxacin
 - [D] Penicillins

14. _____ isomer of Chloramphenicol is clinically active.
[A] D-erythro
[B] L-erythro
[C] D-threo
[D] L-threo
15. Which is not true in case of penicillin?
[A] Good oral absorption but relatively acid labile
[B] Ineffective against Gm –ve bacilli
[C] Useful against Gm+ve cocci
[D] Highly stable to acid /base
16. The naturally occurring tetracyclines contains which of the followings?
[A] α -C4 dimethyl amino substituent
[B] α -C3 dimethyl amino substituent
[C] α -C3 Dihydroxy substituent
[D] α -C3-C4 keto enol group
17. Which of the following is not present in macrolide?
[A] A glycosidically linked amino sugar
[B] A spiroketal group
[C] A large lactone ring
[D] A ketone group
18. Grey baby syndrome is side effect of _____
[A] Chloramphenicol
[B] Tetracycline
[C] Fosfomycin
[D] Vancomycin
19. All are correct about Green Chemistry. Except One
[A] Waste Minimisation at Source
[B] Use of Catalysts in place of Reagents
[C] Using Non-Toxic Reagents
[D] Maximum Use of Organic Solvent
20. Which is following is not a prodrug
[A] Proguanil
[B] Sulfasalazine
[C] Prontosil red
[D] Trimethoprim

SECTION – II

- Q 2 Attempt any TWO of the following**
- A Describe SAR of Fluoroquinolones. 05
- B Write a detail note on macrolide antibiotics. 05
- C Describe SAR of cephalosporins. 05
- Q 3 Attempt any TWO of the following**
- What is the difference between first line of drug and second line of drugs used in tuberculosis? Give name and mechanism of action of any two drugs. 05
- A 05
- B Give the classification of antifungal agents with mechanism of each class. 05
- C What is the different between RNA virus and DNA virus? Write a note on Protease Inhibitors 05

SECTION – III

- Q 4 Attempt any FOUR of the following**
- A What are alkylating agents? Write mechanism of action, synthesis and uses of Chlorambucil. 05
- B Discuss SAR of tetracyclines. 05
- C Classify antineoplastic agents with their chemical structures. (atleast one structure from each class). 05
- D Draw a synthetic pathway for sulphathiazole and ampicillin.. 05
- E List out various antiseptic and disinfectant with their structure. 05
- F What is the rationale behind the design of antimetabolite as anticancer agents? Explain in detail with examples. 05
- Q 5 Attempt any FOUR of the following**
- A What is green chemistry approach? Explain its principles. 05
- B Write a mechanism of action and SAR of chloramphenicol in details. 05
- C Explain SAR of sulphonamides. 05
- D What is immunosuppressive agent? Enlist any four compounds with their structure. 05
- E Give chemical classification of antiamoebic agents with chemical structure from each class. 05
- F What is blood schizonticide? Give the IUPAC name and synthesis of chloroquine. 05
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